Exam Date & Time: 20-Feb-2026 (01:15 PM - 03:30 PM)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Computer Science and Applications
University Theory Examination February 2026
1st Semester of Master of Science in Information Technology (M.Sc.(IT))

ADVANCED DATABASE ADMINISTRATION [CTUC507]

Marks: 50 **Duration: 135 mins.**

Section - I

Answer all the questions. Section Duration: 20 mins

1 Which of the following correctly matches Oracle's memorystructures?

1)	SGA – User-specific memory, PGA – Shared memory for all sessions		2)	SGA – Shared memory, PGA – Memory allocated to a single user session		3)	SGA – Physical data storage, PGA – Logical schema storage		4)	SGA – Backup memory, PGA – Temporary storage for redo logs	

(1)

Course Outcome: "CO1" / Cognitive Level: "Application"

2 Which datatype is most appropriate for storing values like123.45?

1)	INTEGER		2)	FLOAT / REAL		3)	CHAR		4)	BOOLEAN	

(1)

Course Outcome: "CO1" / Cognitive Level: "Application"

3 You need to load data from a text file into an Oracletable.
Which utility should you use?

1)	Data Pump		2)	SQL*Loader		3)	RMAN		4)	SCN Viewer	

(1)

Course Outcome: "CO1" / Cognitive Level: "Application"

4 A DBA wants to improve database performance. Which Oracle memorystructure should be tuned first?

1)	Control Files		2)	SGA		3)	Redo Logs		4)	Data Files	

(1)

Course Outcome: "CO1" / Cognitive Level: "Application"

5 A DBA wants to allow a user to create tables. Which privilegeshould be granted?

1)	CREATE USER		2)	CREATE TABLE		3)	ALTER USER		4)	DROP TABLE	

(1)

Course Outcome: "CO4" / Cognitive Level: "Application"

6 A DBA is designing a secure role-based access model. Which objectshould be created first?

1)	User		2)	Table		3)	Role		4)	Index	

(1)

Course Outcome: "CO5" / Cognitive Level: "Create"

7 A DBA wants to reduce query execution time. Which technique shouldbe applied first? (1)

1)	Backup		2)	Indexing		3)	Auditing		4)	Encryption	

Course Outcome: "CO4" / Cognitive Level: "Application"

8 A DBA wants to protect credit card data. Which technique should beapplied?

1)	Indexing		2)	Encryption		3)	Normalization		4)	Partitioning	

(1)

Course Outcome: "CO4" / Cognitive Level: "Application"

9 A system crash occurred. Which backup type allows full recovery?

1)	Logical backup		2)	Cold backup		3)	Incremental backup		4)	Full backup with archive logs	

(1)

Course Outcome: "CO5" / Cognitive Level: "Application"

10 A DBA is planning migration from MySQL to Oracle. Which Oracleutility should be designed in the migration plan?

1)	RMAN		2)	SQL*Plus		3)	Data Pump (IMPDP/EXPDP)		4)	PL/SQL	

(1)

Course Outcome: "CO5" / Cognitive Level: "Create"

Section-II

Answer 4 out of 6 questions.

1 Analyze the roles andresponsibilities of a DBA in managing a large enterprise databaseenvironment.

(5)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

2 Evaluate the importance of Oracle memory and process structures inimproving database performance.

(5)

Course Outcome: "CO1" / Cognitive Level: "Evaluate"

3 Apply indexing techniques to improve query performance in a largestudent database.

(5)

Course Outcome: "CO2" / Cognitive Level: "Application"

4 Break down the Oracle memory architecture.
Analyze the role of SGA components such as Shared Pool, DatabaseBuffer Cache, and Redo Log Buffer.

(5)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

5 Demonstrate how administrative privileges can be granted andrevoked for different database users in an organization.

(5)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

6 Evaluate different administrative user authentication methods andrecommend the most secure approach.

(5)

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

Section-III

Answer 4 out of 6 questions.

1			Analyze how Virtual Private Database (VPD) enhances data securityin multi-user environments. (5) Course Outcome: "CO3" / Cognitive Level: "Analysis"
2			Discuss common causes of database data loss and explain how backupand recovery can help minimize their impact. (5) Course Outcome: "CO3" / Cognitive Level: "Analysis"
3			Define database backup and explain why it is essential for databasemanagement. (5) Course Outcome: "CO4" / Cognitive Level: "Create"
4			Design a data migration plan for transferring data from a legacysystem to Oracle. (5) Course Outcome: "CO4" / Cognitive Level: "Create"
5			Develop a comprehensive backup and recovery strategy for a hospitaldatabase system. (5) Course Outcome: "CO5" / Cognitive Level: "Create"
6			Evaluate the advantages and disadvantages of distributed databasesystems compared to centralized systems. (5) Course Outcome: "CO5" / Cognitive Level: "Evaluate"

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Event Name	Subject
University Exam Dec-Feb 2026	CTUC507:ADVANCED DATABASE ADMINISTRATION

#	Course Outcome	Description	Marks
1	CO1	Create and manage databases	14
2	CO2	Be familiar with database storage management	15
3	CO3	Manage backup and recovery	15
4	CO4	Control user security	13
5	CO5	Managing database query optimization	13
		Total	**70**

* NA - CO Not Selected

Event Name	Subject
University Exam Dec-Feb 2026	CTUC507:ADVANCED DATABASE ADMINISTRATION

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	9	13(18.57%)
4	Analysis	5	25(35.71%)
5	Create	5	17(24.29%)

6	Evaluate	3	15(21.43%)
	Total	22	70 (100%)